

USDnet++: Distilling Signal Processing Based Dereverberation for Unsupervised Neural Speech Dereverberation

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Abstract

Given a set of unlabeled reverberant speech signals, the recent USDnet model can learn to effectively perform unsupervised single-speaker speech dereverberation. This paper proposes USDnet++, which extends USDnet by leveraging signal processing based dereverberation results to improve the training of USDnet. The rationale is that the dereverberation result produced by signal processing algorithms typically has a higher signal-to-noise (SNR) of the target speech and hence could be leveraged as a weak supervision to improve the training of USDnet. In this paper, we employ weighted prediction error (WPE) and convolutional beamforming (WPD) to derive the signal processing based dereverberation result for improving USDnet. Evaluation results on the WSJ0CAM-DEREVERB dataset demonstrate the effectiveness of USDnet++.

Index Terms: Unsupervised speech dereverberation

1. Introduction

Dereverberation is a crucial and meaningful task, due to the negative impact of room reverberation on speech quality, intelligibility, and the performance of applications like automatic speech recognition (ASR) and speaker identification [1, 2]. To address this issue, many methods have been proposed, ranging from conventional signal processing techniques [3, 4] to deep neural network (DNN) based approaches [5–9].

Conventional dereverberation methods often estimate a linear inverse filter to process the reverberant mixture [1, 10, 11]. A representative algorithm in this category is weighted prediction error (WPE) [3, 12, 13], which suppresses late reverberation while preserving direct sound and early reflections. In multichannel scenarios, beamforming is also widely used, with the MVDR beamformer [11] being a classical example. Recent studies combine these methods, either in a cascade fashion [14, 15] or within joint optimization frameworks [4, 16–18]. Among them, the convolutional beamformer (WPD) [4] achieves strong dereverberation and noise suppression by jointly optimizing spatial and temporal filtering.

With the development of deep learning, deep neural networks (DNNs) have also been applied to speech dereverberation and have shown significant improvements over conventional signal processing based methods. Supervised learning, which trains the DNN on pairs of reverberant and clean speech, has become a widely adopted paradigm in this field [7, 19, 20]. These pairs are typically generated through simulation, which cannot simulate sufficiently realistic signals, and collecting such paired data from real-world recordings is often challenging, even impossible. As a result, DNNs trained purely on simu-

lated data struggle to generalize well to real-recorded signals, limiting their effectiveness in practical applications [21].

Unsupervised learning [22–25] enables training directly on unlabeled real mixtures, without requiring clean references. Some prior studies [26–28] leverage outputs of conventional algorithms for DNN training, such as using DNN-supported WPE results as the training targets [28]. However, these methods are limited by the accuracy of the conventional algorithms.

To overcome this limitation, USDnet [22] avoids directly using the outputs of conventional algorithms as the training targets. Instead, it enforces a reconstruction constraint. That is, the DNN output, after passing through a linear filter, should reproduce the original mixture. Signal processing-based dereverberation (SPD) algorithms, such as WPE [3] and WPD [4], can usually provide a reliable enhancement of the target speech and increase the SNR of the target speech by several decibels. Intuitively, SPD results could be utilized as weak supervision to train DNNs for dereverberation. With this idea in mind, we propose USDnet++, which further incorporates SPD results as additional guidance for model training. Besides encouraging the linearly-filtered DNN estimates to approximate observed mixtures, USDnet++ further encourages them to approximate SPD results. Evaluation results on the WSJ0CAM-DEREVERB dataset [8] show the effectiveness of USDnet++. The contributions of this paper can be summarized as follows:

1. We refine USDnet by adding a new loss term that encourages the filtered DNN estimates to reconstruct the SPD results.
2. We develop a multi-stage refinement framework for USDnet++, where DNN-based and signal-processing-based dereverberation are applied alternately to improve each other.
3. We further enhance USDnet++ by including SPD results as additional inputs to a re-trained DNN, enabling the model to exploit information afforded by SPD to improve dereverb.

2. Physical Model and Objectives

In a reverberant environment with a P -microphone array, a single speaker and weak noises, the recorded signal at a reference microphone $q \in \{1, \dots, P\}$ can be modeled as

$$\begin{aligned} Y_q(t, f) &= X_q(t, f) + V_q(t, f) \\ &= S_q(t, f) + R_q(t, f) + V_q(t, f), \end{aligned} \quad (1)$$

where t indexes T frames and f indexes F frequency bins. Y_q , X_q and V_q are respectively the STFT spectrograms of the observed mixture, reverberant speech, and noise. The reverberant speech X_q is decomposed into the direct-path signal S_q and reverberation R_q . This paper assumes that we have a set of unlabeled, multi-channel reverberant mixtures for model training. Our objective is to use this training set to train unsupervised DNNs that take in the mixture signal at a single microphone or the mixture signals at all the microphones to perform unsupervised speech dereverberation to recover the direct-path signal.

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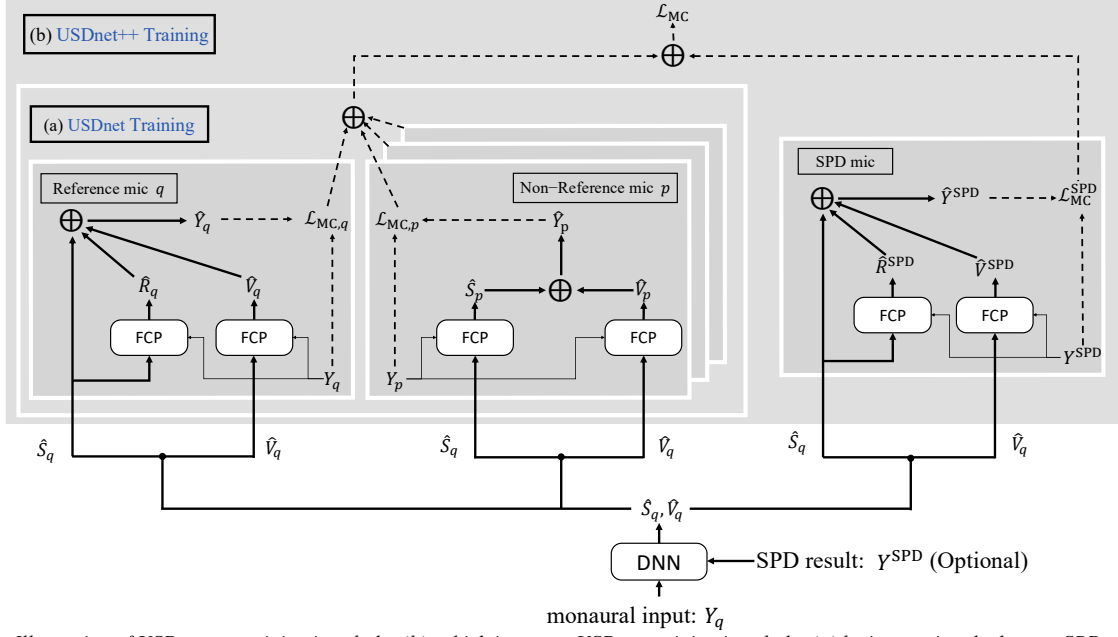


Figure 1: Illustration of USDnet++ training in subplot (b), which improves USDnet training in subplot (a) by integrating the loss on SPD results. FCP is a linear filtering algorithm, which is described in Eq. (5) of Section 3.

3. Review of USDnet

We review USDnet [22] (shown in Fig. 1(a)), on which we build USDnet++. Unsupervised speech dereverberation is a classic blind deconvolution problem [29, 30]. There are infinite solutions to the source and filter pairs, where, in each pair, the source and filter, after linear convolution, can well-approximate the observed reverberant signal, but not respectively corresponding to the true direct-path signal and filter. To address this, USDnet proposes various constraints to regularize its output to have the output approximate the direct-path signal.

USDnet is trained to perform complex spectral masking [31], using the real and imaginary (RI) components of the mixture as input to predict, for the reference microphone q , the RI components of a complex mask for the direct-path signal (denoted as $\hat{M}_q^{(s)}$) and a complex mask for the noise (denoted as $\hat{M}_q^{(v)}$). The two masks are then applied to the mixture to obtain direct-path and noise estimates. That is, $\hat{S}_q = \hat{M}_q^{(s)} \odot Y_q$ and $\hat{V}_q = \hat{M}_q^{(v)} \odot Y_q$, where $\odot$ denotes point-wise multiplication.

USDnet [22] proposes a mixture-constraint (MC) loss $\mathcal{L}_{MC}$ to regularize the two outputs, encouraging $\hat{S}_q$ to approximate the direct-path signal S_q :

$$\mathcal{L}_{MC} = \mathcal{L}_{MC,q} + \alpha \times \sum_{p=1, p \neq q}^P \mathcal{L}_{MC,p} \quad (2)$$

where $\alpha > 0$ is a weighting term for non-reference microphones. The two loss terms, $\mathcal{L}_{MC,q}$ and $\mathcal{L}_{MC,p}$, check whether the two outputs can be linearly filtered to respectively reconstruct the observed mixtures recorded at microphone q and p , thereby promoting unsupervised neural speech dereverberation. They are respectively defined as

$$\begin{aligned} \mathcal{L}_{MC,q} &= \sum_{t,f} \mathcal{F} \left(Y_q(t, f), \hat{Y}_q(t, f) \right) \\ &= \sum_{t,f} \mathcal{F} \left(Y_q(t, f), \hat{S}_q(t, f) + \hat{R}_q(t, f) + \hat{V}_q(t, f) \right) \\ &= \sum_{t,f} \mathcal{F} \left(Y_q(t, f), \right. \\ &\quad \left. \hat{S}_q(t, f) + \hat{\mathbf{g}}_q(f)^H \tilde{\hat{S}}_q(t, f) + \hat{\mathbf{w}}_q(f)^H \tilde{\hat{V}}_q(t, f) \right), \quad (3) \end{aligned}$$

$$\begin{aligned} \mathcal{L}_{MC,p} &= \sum_{t,f} \mathcal{F} \left(Y_p(t, f), \hat{Y}_p(t, f) \right) \\ &= \sum_{t,f} \mathcal{F} \left(Y_p(t, f), \hat{X}_p(t, f) + \hat{V}_p(t, f) \right) \\ &= \sum_{t,f} \mathcal{F} \left(Y_p(t, f), \hat{\mathbf{h}}_p(f)^H \tilde{\hat{S}}_q(t, f) + \hat{\mathbf{w}}_p(f)^H \tilde{\hat{V}}_q(t, f) \right), \quad (4) \end{aligned}$$

where $\tilde{\hat{S}}_q(t, f) = [\hat{S}_q(t - K + 1, f), \dots, \hat{S}_q(t - \Delta, f)]^T \in \mathbb{C}^{K-\Delta}$ and $\tilde{\hat{V}}_q(t, f) = [\hat{V}_q(t - I + 1, f), \dots, \hat{V}_q(t + J, f)]^T \in \mathbb{C}^{I+J}$ stacks a window of time-frequency (T-F) units in $\hat{S}_q$, and $\tilde{\hat{V}}_q(t, f) = [\hat{V}_q(t - L, f), \dots, \hat{V}_q(t + L, f)]^T \in \mathbb{C}^{L+1+L}$ stacks a window of $L + 1 + L$ T-F units in $\hat{V}_q$. $\hat{\mathbf{g}}_q(f) \in \mathbb{C}^{K-\Delta}$, $\hat{\mathbf{h}}_p(f) \in \mathbb{C}^{I+J}$, $\hat{\mathbf{w}}_q(f) \in \mathbb{C}^{L+1+L}$ and $\hat{\mathbf{w}}_p(f) \in \mathbb{C}^{L+1+L}$ denote the linear filters, which need to be estimated and will be described later in the next paragraph. The filter taps, K , Δ , I , J and L , are unknown in unsupervised setup and are treated as tunable hyper-parameters. Their values are fixed and shared across all training utterances. In USDnet, $\mathcal{F}(\cdot, \cdot)$ is defined as $\mathcal{F}(Y_a(t, f), \hat{Y}_a(t, f)) = \frac{1}{\sum_{t',f'} |Y_a(t', f')|} \times \mathcal{G}(Y_a(t, f), \hat{Y}_a(t, f))$, with $\mathcal{G}(\cdot, \cdot)$ defined as $\mathcal{G}(a, b) = |\mathcal{R}(a) - \mathcal{R}(b)| + |\mathcal{I}(a) - \mathcal{I}(b)| + ||a| - |b||$, where $\mathcal{R}(\cdot)$ and $\mathcal{I}(\cdot)$ extract RI components, and $|\cdot|$ computes magnitude.

To compute $\mathcal{L}_{MC}$, the linear filters must be computed first. Each of them can be interpreted as a relative transfer function (RTF) characterizing the relationship between the DNN output and the reverberant signal captured by a microphone. In [22], the relative transfer functions $\hat{\mathbf{g}}_q(f)$, $\hat{\mathbf{h}}_p(f)$, $\hat{\mathbf{w}}_q(f)$ and $\hat{\mathbf{w}}_p(f)$ are computed by forward convolutive prediction (FCP) [27]. For example, $\hat{\mathbf{g}}_q(f)$ can be computed as follows:

$$\hat{\mathbf{g}}_q(f) = \arg \min_{\mathbf{g}_q(f)} \sum_t \frac{|Y_q(t, f) - \mathbf{g}_q(f)^H \tilde{\hat{S}}_q(t, f)|^2}{\hat{\lambda}_q(t, f)}, \quad (5)$$

where $\hat{\lambda}$ is a weighting term defined as $\hat{\lambda}_q(t, f) = \frac{1}{P} \sum_{p=1}^P |Y_p(t, f)|^2 + \xi \times \max(\frac{1}{P} \sum_{p=1}^P |Y_p|^2)$, with ξ (tuned

to 10^{-4}) flooring the weighting term and $\max(\cdot)$ extracting the maximum value of a spectrogram. All the other filters are computed in a similar way.

4. USDnet++

Our proposed USDnet++, shown in Fig. 1(b), improves USDnet in Fig. 1(a) by including new losses defined on dereverberated mixtures computed by conventional SPD algorithms such as WPE [3] and WPD [4]. This section presents the derivation of SPD results and the training procedure of USDnet++, which includes three strategies: (a) SPD-constrained training; (b) multi-stage refinement; and (c) including SPD results as input features.

4.1. SPD-Constrained Training of USDnet++

Given a pre-trained USDnet [22], we first leverage its estimate of the direct-path signal to derive an SPD result (denoted as Y^{SPD}) and then leverage the SPD result to compute an extra loss for fine-tuning USDnet, yielding USDnet++. The procedure for obtaining the SPD result will be detailed later in Section 4.4.

Fig. 1(b) illustrates how USDnet++ leverages the SPD result. The SPD result can be considered as a mixture signal recorded by a virtual microphone. It can be utilized to design a new MC loss term similar to Eq. (3):

$$\begin{aligned}\mathcal{L}_{\text{MC}}^{\text{SPD}} &= \sum_{t,f} \mathcal{F} \left(Y^{\text{SPD}}(t,f), \hat{Y}^{\text{SPD}}(t,f) \right) \\ &= \sum_{t,f} \mathcal{F} \left(Y^{\text{SPD}}(t,f), \right. \\ &\quad \left. \hat{S}_q(t,f) + \hat{\mathbf{g}}^{\text{SPD}}(f)^H \tilde{S}_q(t,f) + \hat{\mathbf{w}}^{\text{SPD}}(f)^H \check{V}_q(t,f) \right),\end{aligned}\quad (6)$$

where $\tilde{S}_q(t,f)$ and $\check{V}_q(t,f)$ are defined in the same way as those in Eq. (3) and the linear filters are computed in the same way as Eq. (5). Building on the loss of USDnet in Eq. (2), the training loss for USDnet++ is defined as

$$\mathcal{L}_{\text{MC}} = \mathcal{L}_{\text{MC},q} + \alpha \times \sum_{p=1, p \neq q}^P \mathcal{L}_{\text{MC},p} + \beta \times \mathcal{L}_{\text{MC}}^{\text{SPD}}, \quad (7)$$

where β denotes a tunable weighting term.

4.2. Multi-stage Refinement for USDnet++

Via SPD-constrained training, USDnet++ is expected to obtain a better estimate of the direct-path signal than USDnet. The better direct-path signal can in turn lead to better SPD results, which can be leveraged to fine-tune the previous USDnet++ model. This procedure can be iterated multiple times to progressively improve SPD results and USDnet++.

4.3. Including SPD Results as Input Features for USDnet++

We further propose a system with two USDnet++ models. After the first USDnet++ model is trained, we use it to compute the direct-path signal and derive an SPD result Y^{SPD} . The SPD result is then used as an input feature in addition to the original mixture to train the second USDnet++ model. Notice that at inference time, the system needs to run the two models to obtain the final dereverberation result.

4.4. Deriving SPD Results

We consider two SPD algorithms, WPE [3] and WPD [4], to obtain SPD results for training USDnet++. Both SPD results

can be computed given a DNN-estimated direct-path signal, $\hat{S}_q$.

4.4.1. WPE Supported by Unsupervised DNN

WPE [3] performs dereverberation in the T-F domain by removing late reverberation estimated by delayed linear prediction. In detail, it formulates dereverberation as an optimization problem aimed at minimizing the weighted prediction error:

$$\arg \min_{\mathbf{w}(f)} \sum_t \frac{|Y_q(t,f) - \mathbf{w}_q(f)^H \bar{\mathbf{Y}}(t,f)|^2}{|\hat{\lambda}_q(t,f)|^2} \quad (8)$$

where $\bar{\mathbf{Y}}(t,f) \in \mathbb{C}^{\Delta_l \times P}$ stacks Δ_l delayed observations across P channels with a prediction delay Δ_d , $\mathbf{w}_q(f)$ is the prediction filter, and $\hat{\lambda}_q$ is a weighting term. $\hat{\lambda}_q$ can be estimated via maximum likelihood estimation by using an iterative algorithm proposed in the original WPE paper [3]. It can also be computed via a DNN-supported solution following the DNN-WPE algorithm [26]. In our study, we leverage the direct-path estimate produced by USDnet or USDnet++ to compute it. That is, $\hat{\lambda}_q(t,f) = |\hat{S}_q(t,f)|$. The WPE result is computed as

$$Y^{\text{SPD}}(t,f) = Y_q(t,f) - \hat{\mathbf{w}}_q(f)^H \bar{\mathbf{Y}}(t,f), \quad (9)$$

which is used for SPD-constrained training.

4.4.2. WPD Supported by Unsupervised DNN

WPD [4] is a popular multi-channel multi-frame filter for speech dereverberation. It usually produces better dereverberation than WPE. It is computed by solving the following problem:

$$\arg \min_{\mathbf{w}_q(f)} \sum_t \frac{|\mathbf{w}_q(f)^H \bar{\mathbf{Y}}(t,f)|^2}{|\hat{\lambda}_q(t,f)|^2}, \text{ s.t. } \mathbf{w}_{q;0}(f)^H \hat{\mathbf{d}}_q(f) = 1, \quad (10)$$

where $\bar{\mathbf{Y}}(t,f) = [\mathbf{Y}(t-\Delta_d-\Delta_l+1,f)^T, \dots, \mathbf{Y}(t-\Delta_d,f)^T, \mathbf{Y}(t,f)^T]^T \in \mathbb{C}^{(\Delta_l+1) \times P}$ with Δ_d denoting a prediction delay and Δ_l the number of filter taps for past frames beyond the prediction delay, $\mathbf{w}_q(f) = [\mathbf{w}_{q;-\Delta_d-\Delta_l+1}(f)^T, \dots, \mathbf{w}_{q;-\Delta_d}(f)^T, \mathbf{w}_{q;0}(f)^T]^T \in \mathbb{C}^{(\Delta_l+1) \times P}$ with $\mathbf{w}_{q;i}(f) \in \mathbb{C}^P$ denoting the filter applied to frame $t+i$ in order to produce the result at the current frame t , and $\hat{\mathbf{d}}_q(f)$ is the estimated relative transfer function for microphone q . Following [32] and based on the DNN-estimated direct-path signal $\hat{S}_q$, $\hat{\lambda}_q$ can be computed as $\hat{\lambda}_q(t,f) = \varepsilon \times \max(|\hat{S}_q|^2 + |\hat{S}_q(t,f)|^2)$, where ε is a floor value to avoid putting too much weight on T-F units with low energy. Through T-F masking and also based on the DNN-estimated target speech $\hat{S}_q$, $\hat{\mathbf{d}}_q(f)$ is computed as the principal eigenvector of an estimated speech covariance matrix:

$$\hat{\Phi}(f) = \sum_t \hat{m}(t,f) \mathbf{Y}(t,f) \mathbf{Y}(t,f)^H, \quad (11)$$

$$\hat{m}(t,f) = \frac{|\hat{S}_q(t,f)|}{|\hat{S}_q(t,f)| + |Y_q(t,f) - \hat{S}_q(t,f)|}, \quad (12)$$

$$\hat{\mathbf{d}}(f) = \mathcal{P}(\hat{\Phi}(f)), \quad (13)$$

$$\hat{\mathbf{d}}_q(f) = \hat{\mathbf{d}}(f) / \hat{d}(f;q), \quad (14)$$

where $\mathcal{P}(\cdot)$ extracts the principal eigenvector, and $\hat{d}(f;q)$ denotes the q^{th} element in $\hat{\mathbf{d}}(f)$. The result of convolutional beamformer is computed as

$$Y^{\text{SPD}}(t,f) = \hat{\mathbf{w}}_q(f)^H \bar{\mathbf{Y}}(t,f), \quad (15)$$

which is used for SPD-constrained training.

Table 1: *Quality of SPD results.*

ID	Method	PESQ $\uparrow$	eSTOI $\uparrow$	SI-SDR(dB) $\uparrow$
0	Mixture	1.64	0.494	-3.6
1	WPE [3]	2.02	0.690	2.0
2a	supDNN-WPE [26]	2.06	0.721	2.7
2b	supDNN-WPD [4]	2.68	0.853	4.4
3a	USDnet-WPE	2.04	0.704	2.1
3b	USDnet-WPD	2.59	0.809	2.8
4a	Oracle-WPE	2.09	0.729	2.8
4b	Oracle-WPD	2.83	0.873	4.9

Table 2: *Results of SPD-constrained training.*

ID	Method	SPD results in $\mathcal{L}_{MC}^{SPD}$	#CH in input/loss	PESQ $\uparrow$	eSTOI $\uparrow$	SI-SDR (dB) $\uparrow$
0	Mixture	-	-	1.64	0.494	-3.6
1a	WPE [3]	-	8 / -	2.02	0.690	2.0
1b	USDnet [22]	-	1 / 8	2.64	0.794	3.1
2a	USDnet++	WPE [3]	1 / 8	2.67	0.797	3.6
2b	USDnet++	USDnet-WPE	1 / 8	2.67	0.798	3.6
2c	USDnet++	USDnet-WPD	1 / 8	2.86	0.822	3.9
3a	USDnet++	Oracle-WPE	1 / 8	2.70	0.811	3.9
3b	USDnet++	Oracle-WPD	1 / 8	2.95	0.849	4.9

5. Experimental Setup

We evaluate USDnet++ on WSJ0CAM-DEREVERB [8], a popular dataset for supervised dereverberation. Based on WSJ0CAM, which provides 7861, 742, and 1088 clean signals for training, validation, and testing, respectively, WSJ0CAM-DEREVERB simulates 39293 (~ 77.7 h), 2968 (~ 5.6 h), and 3262 (~ 6.4 h) reverberant mixtures for training, validation, and testing, respectively. Each utterance is convolved with simulated RIRs generated using random room configurations, speaker-array distances (0.75–2.5m), and T60 values (0.2–1.3 s). An 8-microphone circular array (with a diameter of 20 cm) is used. Diffuse air-conditioning noise randomly sampled from the REVERB dataset [33] is included, with SNRs randomly drawn from the range [5, 25] dB. The sampling rate is 16 kHz.

For STFT, we use a 32 ms window size, an 8 ms hop size, and a square-root Hann analysis window. The DNNs adopt TF-GridNet [8, 9] with hyper-parameters $D = 128$, $B = 4$, $I = 4$, $J = 4$, $H = 200$, $L = 4$, $E = 2$. The batch size is fixed at 2. For complex ratio masking, the predicted RI components are clipped to the range $[-5, 5]$. The values of α and β in Eq. (7) are tuned to 3/8 and 1, respectively.

We compare USDnet++ with two unsupervised baselines: WPE [3] and USDnet [22]. For 8-channel WPE and WPD, we set $\Delta_d = 3$ and $\Delta_l = 5$, following [32]. All methods use the same STFT configuration. The direct-path signal at the reference microphone is used as the reference signal for metric computation. The evaluation metrics include PESQ [34], eSTOI [35], and SI-SDR [36].

6. Evaluation Results

Table 1 presents the scores of the SPD results obtained by (1) the vanilla WPE algorithm [3]; (2) WPE and WPD algorithms supported by supervised DNN, denoted as **supDNN-WPE** and **supDNN-WPD**, where the direct-path signal estimated by supervised DNN (with the same DNN architecture as USDnet++) is used to derive the WPE and WPD filters; and (3) WPE and WPD algorithms supported by unsupervised USDnet, denoted as USDnet-WPE and USDnet-WPD, where the direct-path sig-

Table 3: *Results of multi-stage refinement.*

ID	Method	SPD results in $\mathcal{L}_{MC}^{SPD}$	Stage	PESQ $\uparrow$	eSTOI $\uparrow$	SI-SDR (dB) $\uparrow$
1	USDnet [22]	-	0	2.64	0.794	3.1
2	USDnet++	USDnet-WPD	1	2.86	0.822	3.9
3	USDnet++	USDnet++-WPD	2	2.89	0.826	3.9
4	USDnet++	USDnet++-WPD	3	2.90	0.828	3.9

Table 4: *Results of including SPD results as input features.*

ID	Method	Extra input features	#DNNs	#CH in input/loss	PESQ $\uparrow$	eSTOI $\uparrow$	SI-SDR (dB) $\uparrow$
1	USDnet [22]	-	1	1 / 8	2.64	0.794	3.1
2	USDnet++	USDnet++-WPD	2	8 / 8	2.94	0.840	3.8
3	USDnet++	Oracle-WPD	2	8 / 8	3.14	0.887	5.2

nal estimated by USDnet is used to compute the filters. We also provide oracle WPE and WPD results where the true direct-path signal is used to compute the filters. Although the algorithms can perform dereverberation to some extent, the scores indicate that the processed signals are still far from clean, leaving noticeable residual reverberation and distortions.

Table 2 reports the performance of USDnet++, which fine-tunes a pre-trained USDnet by leveraging SPD results to compute an extra loss. Compared to WPE and USDnet, USDnet++ achieves better dereverberation. Importantly, although USDnet++ leverages the SPD results as weak supervision, its performance surpasses that of SPD results. This can be observed by, for example, comparing system 2a in Table 2 with system 1 in Table 1, and system 2c in Table 2 with system 3b in Table 1. These comparisons indicate that USDnet++ can learn from, and go beyond, SPD.

Table 3 reports the results of multi-stage refinement. By alternating between unsupervised DNN- and WPD-based dereverberation iteratively, each stage achieves progressively better results, confirming the effectiveness of multi-stage refinement.

Table 4 compares the retrained USDnet++ (which incorporates 8-channel SPD results as additional input features). Specifically, the 8-channel WPD results are derived using the USDnet++ model in system 4 of Table 3. The retrained model further boosts metric scores, highlighting the benefit of providing richer input information afforded by SPD to the DNN and demonstrating that combining SPD results with the original mixtures allows USDnet++ to exploit complementary cues for superior dereverberation.

7. Conclusions

We have proposed USDnet++, which leverages signal processing based dereverberation results to improve the training of USDnet. Experiments on WSJ0CAM-DEREVERB show that USDnet++ consistently outperforms conventional algorithms such as WPE and WPD, achieving higher PESQ, eSTOI, and SI-SDR scores. The gains are consistent across evaluation metrics, indicating improvements in perceptual quality, speech intelligibility, and distortion reduction simultaneously. Multi-stage refinement further improves performance by iteratively updating the WPD results and the DNN model, with the WPD outputs used to compute the MC loss. Incorporating SPD results as extra features of USDnet++ enables the network to exploit complementary cues for superior dereverberation. These results confirm the effectiveness of USDnet++. Future research would extend USDnet++ to real-recorded, multi-speaker scenarios.

8. Generative AI Use Disclosure

Generative AI tools were used for language polishing to improve the readability of the manuscript. All technical content, experimental design, analysis, and conclusions were developed by the authors, who take full responsibility for this paper.

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